

SURNAME AND INITIALS: NP ZIQUBU

STUDENT NUMBER: 45369666

**CMPG 325 PROJECT MILESTONE 1 : CLIENT DESIGN
REVIEW**

1. Client Requirements

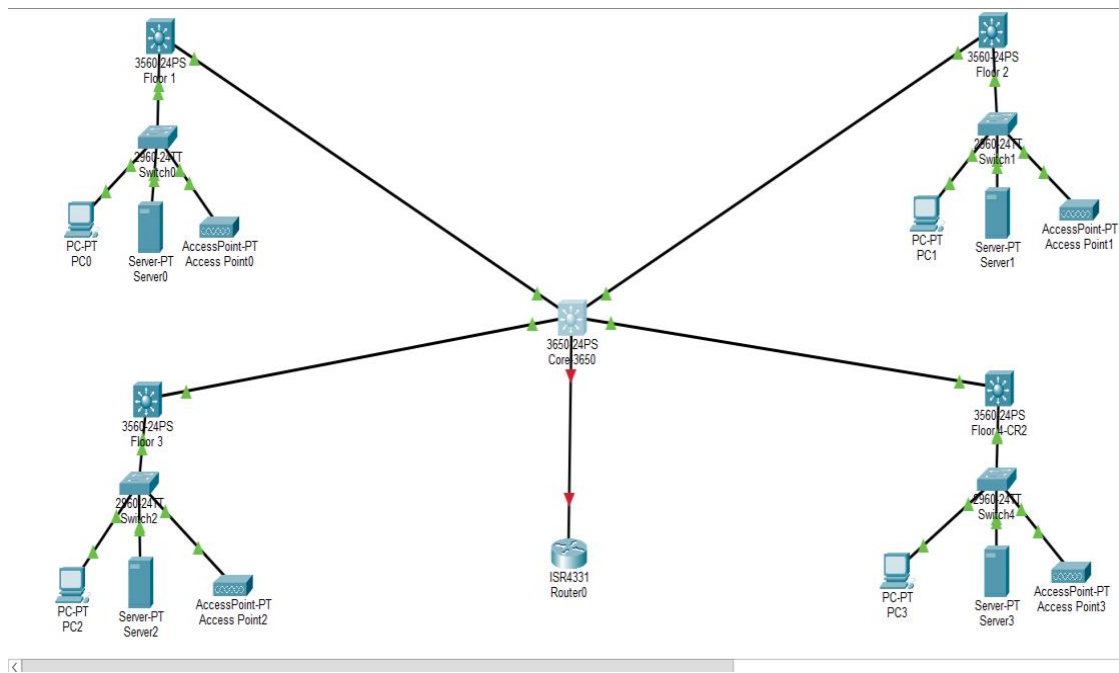
PROBLEM

- Organisation: Kabelo Telecom Installations (Rustenburg)
- Industry: Telecommunications
- Current challenge: No proper network design for new floor/area, guest users risk accessing internal resources.
- Goal: Reliable, secure connectivity with proper segmentation and coverage expansion.
- Networking challenge: Implement Port Security to prevent unauthorized device access.

REQUIREMENTS

- Users: Staff, guest users, servers.
- Departments: Office staff, technical team, guest visitors.
- Applications/Services: Internet access, file sharing, VoIP, DHCP, DNS.
- Security: Guest isolation, port security, VLAN segmentation.
- Budget: Use Cisco Packet Tracer simulation (no hardware cost).
- Future growth: Must accommodate expansion to new floor/area.

2. Physical Topology



- 1x Core Switch - 3650-24PS in the middle - this is the heart
- 4x Floor Distribution Switches - 3560-24PS - one per floor (Floor 1,2,3,4)

- 4x Access Switches - 2960-24TT - one per floor, under the 3560
- 1x Router - ISR4331 - connected to Core for internet/WAN
- End devices per floor: 1 PC, 1 Server, 1 Access Point
- Cables:
 - Core to Floor 3560s = Copper Straight-Through
 - 3560 to 2960 = Copper Straight-Through
 - 2960 to PC/Server/AP = Copper Straight Through

3. Logical Topology

Your logical topology is a VLAN-Segmented Logical Bus over a Physical Extended Star, sub netted from 10.55.0.0/16.

1. IP Subnetting / VLAN Mapping:
 - 10.55.1.0/24 (Staff VLAN): Range 10.55.1.1 - 10.55.1.254. Gateway 10.55.1.1 on Core-3650. For all staff PCs across floors.
 - 10.55.2.0/24 (Guest VLAN): Range 10.55.2.1 - 10.55.2.254. Gateway 10.55.2.1. For guest Wi-Fi / APs. Isolated from Staff/Server for security.
 - 10.55.3.0/24 (Server VLAN): Range 10.55.3.1 - 10.55.3.254. Gateway 10.55.3.1. For all servers.
 - 10.55.4.0/24 (New Floor Staff VLAN): Range 10.55.4.1 - 10.55.4.254. Gateway 10.55.4.1. For expansion Floor 4 staff, proving scalability.
2. Logical Operation:
 - Trunk Links: All uplinks (Core to Floor 3560, Floor 3560 to Access 2960) are 802.1Q trunks carrying all 4 VLANs at once. One physical cable = 4 logical networks.
 - Access Links: Downlinks to PCs/Servers/APs are access ports assigned to ONE VLAN only. e.g. switch port access VLAN 1 for Staff.
 - Inter-VLAN Routing: Core-3650 does the routing. It has 4 SVIs, one per subnet above. Without it, 10.55.1.0 cannot talk to 10.55.3.0.
 - DHCP: Core-3650 will have 4 DHCP pools, one per subnet, handing out IPs from the ranges in your table with default-router set to the gateway.
 - Traffic Flow: Logical communication is not star. A Staff PC (10.55.1.10) talking to another Staff PC (10.55.1.20) on a different floor goes logically direct via Core, but stays in same VLAN. Staff to Server traffic must hop from VLAN 1 to VLAN 3 via Core SVI.

4. IP Addressing plan

Subnet	CIDR	Range	Purpose
10.55.1.0/24	/24	10.55.1.1 – 10.55.1.254	Staff VLAN
10.55.2.0/24	/24	10.55.2.1 – 10.55.2.254	Guest VLAN
10.55.3.0/24	/24	10.55.3.1 – 10.55.3.254	Server VLAN

10.55.4.0/24	/24	10.55.4.1 – 10.55.4.254	New floor staff VLAN
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5. **Initial GitHub repository**

<https://github.com/Nokuthula-22/CMPG-325-PROJECT--Kabelo-Telecom-Installations>